Reverse-check review package — `Top-Philic-Zprime_gen`

An independent, **blank-slate** agent instance reconstructed the physics below from the sanitized `.fr` alone (no paper, no metadata, no conversation history). A second fresh instance then compared the reconstruction against the source paper. The final verdict belongs to the human reviewer — sign off at the bottom.

item	value
model file	`Top-Philic-Zprime/model/Top-Philic-Zprime_gen.fr`
original model name	`Top-Philic-Zprime_gen` (hidden from the agent)
paper	Top-Philic-Zprime/text/1604.07421.txt
blindness scope	blank slate w.r.t. paper, authors, model name, comments and prose labels — NOT w.r.t. field/parameter symbol names, which are kept so the reviewer can map the reconstruction back

Verbatim Lagrangian terms (from the `.fr`)

These are the terms the reconstruction must account for, quoted unmodified. Check each against its reconstructed LaTeX form below.

LV1kin (=)

```
-1/4 (del[V1[nu], mu] - del[V1[mu], nu]) (del[V1[nu], mu] - del[V1[mu], nu]) + 1/2 MV1^2
↪  V1[mu] V1[mu]
```

LV1intL (=)

```
Block[{sp1, sp2, cc}, cL V1[mu] QLbar[sp1, 1, 3, cc] Ga[mu, sp1, sp2] QL[sp2, 1, 3, cc]]
```

LV1intR (=)

```
Block[{sp1, sp2, cc}, cR V1[mu] uRbar[sp1, 3, cc] Ga[mu, sp1, sp2] uR[sp2, 3, cc]]
```

LV1int (=)

```
LV1intL + LV1intR
```

LBSM (=)

```
LV1kin + LV1int
```

Blank-slate reconstruction

Reconstructed Physics From `sanitized.fr`

Lagrangian

The model contains one new real neutral massive vector field $V_{1\mu}$. It has no declared nontrivial gauge indices and has

$$Q(V_1) = 0, \qquad Y(V_1) = 0.$$

Since V_1 is a gauge singlet, its covariant derivative is just

$$D_\mu V_{1\nu} = \partial_\mu V_{1\nu}.$$

The corresponding Abelian field strength appearing in the file is

$$V_{1\mu\nu} \equiv \partial_\mu V_{1\nu} - \partial_\nu V_{1\mu}.$$

LV1kin

```
LV1kin = -1/4 (del[V1[nu], mu] - del[V1[mu], nu])
              (del[V1[nu], mu] - del[V1[mu], nu])
         + 1/2 MV1^2 V1[mu] V1[mu];
```

$$\mathcal{L}_{\texttt{LV1kin}} = -\frac{1}{4} V_{1\mu\nu} V_1^{\mu\nu} + \frac{1}{2} M_{V_1}^2 V_{1\mu} V_1^{\mu}.$$

LV1intL

```
LV1intL =
Block[{sp1, sp2, cc},
  cL V1[mu] QLbar[sp1, 1, 3, cc]
     Ga[mu, sp1, sp2]
     QL[sp2, 1, 3, cc]
];
```

Here `QL[sp, 1, 3, cc]` is the upper component of the third-generation left-handed quark doublet, i.e. the left-handed top quark t_L, with color index c. The color index is contracted diagonally.

$$\mathcal{L}_{\texttt{LV1intL}} = c_L \, V_{1\mu} \, \bar{t}_{L,c} \gamma^\mu t_L^c.$$

Equivalently, using chiral projectors,

$$\mathcal{L}_{\texttt{LV1intL}} = c_L \, V_{1\mu} \, \bar{t}_c \gamma^\mu P_L t^c, \qquad P_L = \frac{1-\gamma^5}{2}.$$

LV1intR

```
LV1intR =
Block[{sp1, sp2, cc},
  cR V1[mu] uRbar[sp1, 3, cc]
     Ga[mu, sp1, sp2]
     uR[sp2, 3, cc]
];
```

Here `uR[sp, 3, cc]` is the right-handed top quark t_R, again with a diagonally contracted color index.

$$\mathcal{L}_{\texttt{LV1intR}} = c_R \, V_{1\mu} \, \bar{t}_{R,c} \gamma^\mu t_R^c.$$

Equivalently,

$$\mathcal{L}_{\texttt{LV1intR}} = c_R \, V_{1\mu} \, \bar{t}_c \gamma^\mu P_R t^c, \qquad P_R = \frac{1+\gamma^5}{2}.$$

LV1int

```
LV1int = LV1intL + LV1intR;
```

$$\mathcal{L}_{\texttt{LV1int}} = V_{1\mu} \, \bar{t}_c \gamma^\mu \left(c_L P_L + c_R P_R \right) t^c.$$

LBSM

```
LBSM = LV1kin + LV1int;
```

$$\mathcal{L}_{\texttt{LBSM}} = -\frac{1}{4} V_{1\mu\nu} V_1^{\mu\nu} + \frac{1}{2} M_{V_1}^2 V_{1\mu} V_1^{\mu} + V_{1\mu}\, \bar{t}_c \gamma^\mu \left(c_L P_L + c_R P_R\right) t^c.$$

The internal coupling definitions are

$$c_L = c_t \cos\theta, \qquad c_R = c_t \sin\theta,$$

where the `.fr` symbols are `cL`, `cR`, `ct`, and `thetaV1`.

Thus the interaction can also be written as

$$\mathcal{L}_{\texttt{LV1int}} = c_t\, V_{1\mu}\, \bar{t}_c \gamma^\mu \left(\cos\theta\, P_L + \sin\theta\, P_R\right) t^c.$$

Field Table

`.fr` symbol	Particle	Spin	SU(3) rep	SU(2) rep	$U(1)_Y$	Electric charge	Self-conjugate	Mass
`V1`	$V_{1\mu}$	1	singlet, no color index declared	singlet, no weak index declared	0	0	yes	`MV1 = 1500.`

The vector width is internal:

$$\Gamma_{V_1} = \texttt{WV1}.$$

For $M_{V_1} > 2m_t$, the file defines

$$\Gamma_{V_1} = \frac{c_t^2 M_{V_1}}{8\pi} \sqrt{1 - \frac{4m_t^2}{M_{V_1}^2}} \left[1 - \frac{m_t^2}{M_{V_1}^2} \left(1 - 3\sin 2\theta\right)\right],$$

and otherwise

$$\Gamma_{V_1} = 0.$$

Parameters

`.fr` symbol	Type	Value	Appears in	Physical meaning
`ct`	external	`2.`	`cL`, `cR`, `WV1`	Overall strength of the $V_1\bar{t}t$ coupling
`thetaV1`	external	`1.5707963267948966`	`cL`, `cR`, `WV1`	Chiral mixing angle controlling the relative left- and right-handed top couplings
`cL`	internal	$c_t \cos\theta$	`LV1intL`	Left-handed top coupling
`cR`	internal	$c_t \sin\theta$	`LV1intR`	Right-handed top coupling
`WV1`	internal	width formula above	particle width of `V1`	Total width for $V_1 \to t\bar{t}$, when kinematically open

Only `ct` and `thetaV1` are new external parameters in the file.

Physics Summary

This model adds a single real, electrically neutral, color-singlet massive vector boson V_1 with direct vector-current interactions only to top quarks. Its couplings are chiral, with independent left- and right-handed pieces fixed by $c_L = c_t \cos\theta$ and $c_R = c_t \sin\theta$.

It mediates processes involving top-quark pairs, especially resonant or off-shell production and decay $V_1 \leftrightarrow t\bar{t}$, with chirality-dependent amplitudes.

Paper cross-check

Comparison of `reconstruction.md` Against the Paper Model

Paper Model Location

The model is defined in **Section 2, "A Top-Philic Resonance: Simplified Model"**, especially **Section 2.1, "Setup"**. The paper introduces a **color-singlet vector particle** V_1 that dominantly couples to $t\bar{t}$, and gives the relevant interaction Lagrangian in **Eq. (2.1)**:

$$\mathcal{L}_{\rm int} = \bar{t}\gamma_\mu (c_L P_L + c_R P_R) t\, V_1^\mu = c_t \bar{t}\gamma_\mu(\cos\theta P_L + \sin\theta P_R) t\, V_1^\mu.$$

The chiral projectors and coupling definitions are stated immediately below Eq. (2.1):

$$P_{R/L} = (1 \pm \gamma^5)/2, \qquad c_t = \sqrt{c_L^2 + c_R^2}, \qquad \tan\theta = c_R/c_L.$$

The decay width is given in **Eq. (2.2)**:

$$\Gamma(V_1 \to t\bar{t}) = \frac{c_t^2 M_{V_1}}{8\pi}\sqrt{1 - \frac{4m_t^2}{M_{V_1}^2}}\left[1 - \frac{m_t^2}{M_{V_1}^2}\left(1 - 3\sin(2\theta)\right)\right].$$

Section 2.2 states that the paper treats (M_{V_1}, c_t, θ) as the three free parameters and then sets $\theta = \pi/2$ for the rest of the collider study. Section 3 says the MadGraph implementation is based on Eq. (2.1) with parameters M_{V_1}, c_t, θ.

Term-by-Term Comparison

paper term (eq. ref)	reconstruction term	verdict	notes
Color-singlet vector particle V_1, Section 2.1	One new real neutral massive vector field $V_{1\mu}$, color singlet, weak singlet, $Q = Y = 0$, self-conjugate	agree	The paper explicitly says "color singlet vector particle." It does not explicitly list $SU(2)_L$, $U(1)_Y$, electric charge, or self-conjugacy, but the neutral $V_1^\mu \bar{t}\gamma_\mu t$ interaction implies no electric or color charge. Weak-singlet status is an implementation-level completion of the simplified post-EWSB interaction.
$\bar{t}\gamma_\mu c_L P_L t\, V_1^\mu$, Eq. (2.1)	$c_L V_{1\mu} \bar{t}_c \gamma^\mu P_L t^c$, reconstructed from `QLbar[...,1,3,cc]` `QL[...,1,3,cc]`	agree	Physics content matches: left-handed top current only, diagonal color contraction, no bottom or light-quark current.

paper term (eq. ref)	reconstruction term	verdict	notes
$\bar{t}\gamma_\mu c_R P_R t\, V_1^\mu$, Eq. (2.1)	$c_R V_{1\mu} \bar{t}_c \gamma^\mu P_R t^c$, reconstructed from `uRbar[...,3,cc]` `uR[...,3,cc]`	agree	Physics content matches: right-handed top current only, diagonal color contraction.
Full interaction $\bar{t}\gamma_\mu(c_L P_L + c_R P_R)tV_1^\mu$, Eq. (2.1)	$V_{1\mu}\bar{t}_c\gamma^\mu(c_L P_L + c_R P_R)t^c$	agree	Same chiral current structure. Differences are only index placement and metric/notational conventions.
Reparametrized interaction $c_t\bar{t}\gamma_\mu(\cos\theta P_L + \sin\theta P_R)tV_1^\mu$, Eq. (2.1)	$c_t V_{1\mu}\bar{t}_c\gamma^\mu(\cos\theta P_L + \sin\theta P_R)t^c$	agree	Matches the paper's second form of Eq. (2.1).
$P_{R/L} = (1 \pm \gamma^5)/2$, text below Eq. (2.1)	$P_L = (1-\gamma^5)/2$, $P_R = (1+\gamma^5)/2$	agree	Exact match.
$c_t = \sqrt{c_L^2 + c_R^2}$, $\tan\theta = c_R/c_L$, text below Eq. (2.1)	$c_L = c_t\cos\theta$, $c_R = c_t\sin\theta$	agree	Equivalent when c_t is taken positive and θ carries the quadrant/sign convention. A human should note that the reconstruction uses the implementation's explicit parametrization rather than the paper's inverse definitions.
$\Gamma(V_1 \to t\bar{t})$, Eq. (2.2)	Same expression for Γ_{V_1}, with $\Gamma_{V_1} = 0$ below threshold	agree	The formula matches Eq. (2.2). The below-threshold zero is an implementation detail; the paper focuses on TeV-range M_{V_1} and only remarks that other modes below $2m_t$ are possible.
Three free parameters (M_{V_1}, c_t, θ), Section 2.2 and Section 3	Parameters `MV1`, `ct`, `thetaV1`; defaults `MV1=1500`, `ct=2`, `thetaV1=pi/2`	agree	The parameter set matches. The numerical values correspond to the paper's benchmark and later choice $\theta = \pi/2$, but the paper scans M_{V_1} and c_t; the defaults should not be read as the full paper model.
Paper later sets $\theta = \pi/2$, Section 2.2	`thetaV1 = 1.5707963267948966`	agree	This matches the right-handed-top benchmark used for the rest of the paper.

paper term (eq. ref)	reconstruction term	verdict	notes
Contact interaction after integrating out V_1 at $\theta = \pi/2$: $\frac{1}{2}\frac{c_t^2}{M_{V_1}^2}(\bar{t}_R\gamma_\mu t_R)(\bar{t}_R\gamma^\mu t_R)$, Eq. (2.3)	No contact operator listed	missing-in-reconstruction	This is not part of the fundamental implemented Lagrangian; it is an effective comparison used for ATLAS bounds. Missing it is not a mismatch for the UV/simplified model implementation.
No explicit kinetic or Proca mass term written in the paper's model equations	$-\frac{1}{4}V_{1\mu\nu}V_1^{\mu\nu}+\frac{1}{2}M_{V_1}^2 V_{1\mu}V_1^\mu$	extra-in-reconstruction	The paper gives only the "relevant interaction" Lagrangian in Eq. (2.1), not the full free-field Lagrangian. The kinetic and mass terms are standard for a massive real vector and consistent with the stated massive resonance.
Paper assumes other interactions are weak/negligible, Section 2.1	Reconstruction includes only $V_1\bar{t}t$ interactions	agree	The reconstruction correctly omits light-quark, bottom, lepton, gauge-boson, and Higgs interactions.
Possible below-threshold modes for $M_{V_1} < 2m_t$, mentioned in Section 2.2 via Ref. [37]	Reconstruction sets $\Gamma_{V_1} = 0$ below $2m_t$	disagree	For the paper's TeV-range study this is irrelevant, but strictly the paper acknowledges other possible decay modes below threshold rather than defining a zero total width there.

Disagreements and Checks

- **Extra Proca kinetic/mass term** — severity: **cosmetic**. A human should check whether the review target is the paper's displayed interaction-only Lagrangian or the full FeynRules implementation, because the extra term is standard and physically expected for event generation.
- **Below-threshold width set to zero** — severity: **convention**. A human should check whether the implementation is intended only for $M_{V_1} > 2m_t$, since the paper explicitly points to possible sub-threshold decays but does not use them in the TeV analysis.
- **Contact operator Eq. (2.3) absent** — severity: **cosmetic**. A human should check whether the reconstruction is meant to include derived EFT limits; it is not necessary for reproducing the simplified model Lagrangian Eq. (2.1).
- **Default numerical values $M_{V_1} = 1.5$ TeV, $c_t = 2$, $\theta = \pi/2$** — severity: **convention**. A human should check that these are treated as benchmark/default parameter-card values, not as fixed model definitions, because the paper scans M_{V_1} and c_t.

Overall Assessment

The reconstruction captures the core model defined by the paper: a color-singlet neutral vector resonance V_1 with chiral couplings only to the top quark, parametrized by c_t and θ, with the same $V_1 \to t\bar{t}$ width as Eq. (2.2). The main differences are about scope rather than the central interaction: the reconstruction includes the standard free massive-vector terms that the paper does not explicitly write, omits the derived contact operator used only for bounds, and encodes a zero below-threshold width despite the paper noting possible non-$t\bar{t}$ decays in that region. For the TeV-scale four-top simplified model studied in the paper, the physics content is closely aligned.

Suggested checks for the reviewer

1. Every verbatim `.fr` term above has a reconstructed LaTeX counterpart with the same field content, chirality and conjugation.
2. Kinetic terms: covariant derivative gauge content matches the field's representations; normalization is canonical.
3. Non-self-conjugate interaction terms appear together with their Hermitian conjugates.
4. Quantum numbers in the field table match the `.fr` declarations (and the paper, where the cross-check table flags disagreements).
5. Numeric masses/couplings are placeholders unless the paper pins them — treat values as demo inputs, not measurements.
6. Sanitizer scope: 6 prose labels were scrubbed; field/parameter symbol names were kept and may hint at the model's identity.

Physicist sign-off

- Reviewed by: ________________________________ Date: _________________
- Verdict: [ ] approve [ ] approve with corrections [ ] reject
- Notes: